Financing Firms

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Abstract

Evidence shows that labour and financial markets are closely linked, and shocks are rapidly transmitted between them. Using a model with labour frictions and typical financial frictions, such as costly dividend adjustment and cash flow-based borrowing constraints, this paper shows that the connection between the two markets operates through the firm's marginal value of internal funds. This paper also demonstrates that through the firm's marginal value of internal funds, a low fundamental surplus can amplify shock transmission between the two markets.

1 Introduction

Financial and labour markets are closely connected. Periods of financial distress often bring sharp hiring contractions and rising unemployment. At the same time, improvements in labour market conditions are also often associated with higher firm valuations and easier access to external finance. These patterns suggest that labour market outcomes and firms' financial decisions are jointly determined. Much of the literature, however, studies labour market frictions and financial frictions separately, leaving the

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mechanisms linking the two markets only partially understood.

This paper argues that labour and financial market connection operates through the firm's marginal value of internal funds, which can explain the observed co-movement between financial and labour markets. If a firm is under no financial frictions, the marginal value of internal funds is equal to exactly one. In this case, the financial market plays little role in firms' hiring decisions beyond current profitability. Under financial frictions, particularly costly dividend adjustment and a cash flow-based borrowing constraint, the marginal value of internal funds exceeds unity and varies with the degree of financial stress. This is because additional funds let firms lower the costs of dividend adjustments and relax borrowing constraints. As a consequence, hiring, investment and financing decisions become linked through a common set of shadow values. Movements in financial conditions therefore affect vacancy creation and employment, while labour-market developments influence firms' financial choices.

This mechanism requires dividend adjustment costs and a cash flow-based borrowing constraint to affect the firm simultaneously. Costly dividend adjustment makes the firm's value of internal funds dependent on the dividends it can pay, while the cash-flow-based borrowing constraint makes internal funds scarce; together, these effects make the firm's hiring, investment, and financing decisions sensitive to shocks. Neither friction alone generates quantitatively important interactions between labour and financial markets.

In addition, this mechanism requires the borrowing constraint to be a cash-flow-based constraint tied to the firm's Earnings Before Interest, Taxes, Depreciation and Amortisation (EBITDA), which many firms' lending practices are empirically argued to reflect closely (Lian and Ma, 2021). If, alternatively, the borrowing constraint is a conventional

asset-based collateral constraint in which borrowing is limited by the resale value of the firm's capital stock, the mechanism is weak. This is because, with an EBITDA-based borrowing constraint, hiring decisions affect future cash flows and borrowing capacity. Consequently, changes in labour market conditions alter the firms' financial capacity, creating a feedback mechanism between the two markets. In contrast, under an asset-based constraint, the link is weaker because borrowing capacity depends primarily on collateral values.

The marginal value of internal funds also makes the fundamental surplus a key determinant of the strength of the labour-financial market link. This is because a low fundamental surplus reduces the cash flows generated by firm-worker matches and therefore tightens firms' borrowing constraints. This raises the marginal value of internal funds and amplifies the response of both labour-market and financial-market variables to shocks. As a result, labour-market shocks have larger financial consequences and financial shocks have larger employment consequences when the fundamental surplus is low. The fundamental surplus therefore amplifies the transmission of shocks between the two markets.

To justify this argument, we develop a Real Business Cycle Diamond-Mortensen-Pissarides (RBC-DMP) model with Jermann and Quadrini (2012)-style dividend adjustment costs and EBITDA-based borrowing constraints, and two alternatives: one with no borrowing constraints and one with asset-based borrowing constraints. The model first shows that, under EBITDA borrowing constraints, the contemporaneous correlation between financial and labour market variables is closer to the data than under asset-based borrowing constraints. This highlights that jointly modelling dividend adjustment costs and cash-flow-based borrowing constraints is essential to generate

realistic interactions between labour and financial variables. Second, using productivity shocks and shocks to the borrowing constraint, the model shows that only under EBITDA borrowing constraints, both shocks transmit rapidly across financial and labour markets. If the borrowing constraint is asset-based, or if dividend adjustment costs are removed, transmission is weak or disappears entirely. This highlights the role of the marginal value of internal funds in making the firm's hiring, investment, and financing decisions sensitive to shocks. Third, the model with dividend adjustment costs and EBITDA borrowing constraints also shows amplified responses in both markets to productivity and borrowing-constraint shocks under a low fundamental surplus. Without dividend adjustment, however, the low fundamental surplus's amplification of a borrowing-constraint shock disappears. This provides evidence that a low fundamental surplus amplifies the labour-financial market link via the firm's marginal value of internal funds.

Related Literature This paper contributes to the literature that analyses the financial-labour channel through the lens of the Merz (1995); Andolfatto (1996)-style RBC-DMP model. Of this literature Moiseeva (2026) and Garin (2015) are relevant to this paper. Moiseeva (2026) uses a DMP-RBC model with Kiyotaki and Moore (1997) borrowing constraints to explain that the combination of labour and financial frictions results in two-way spillovers between labour and financial markets that amplify shocks. Garin (2015) using a similar model as Moiseeva (2026), also with a Kiyotaki and Moore (1997) borrowing constraint and intra-period bonds in the style of Jermann and Quadrini (2012), identifies similar amplifications following financial and productivity shocks. More generally, Moiseeva (2026) points to the wider consensus in the literature that financial and labour frictions, of various kinds, together amplify aggregate shocks, with which this paper's analysis aligns. The contribution this paper makes to this particular literature is

that the combination of EBITDA borrowing constraints, dividend adjustment costs and labour frictions, establish a two-way labour-financial market channel that is amplified by a low fundamental surplus, following both productivity and borrowing constraint shocks.

This paper is also related to the literature on the empirical relevance of earnings-based borrowing constraints. Lian and Ma (2021) shows that, in the data, asset-based lending accounts for only a fifth of US corporate debt by value for medium to large firms, while cash flow-based borrowing constraints govern the rest. Drechsel (2023) shows that, following investment shocks, earnings-based borrowing constraints are empirically more relevant than asset-based borrowing constraints. This paper contributes to the literature by providing theoretical evidence that dividend adjustment costs and EBITDA borrowing constraints amplify shocks. In particular, these financial frictions significantly amplify the effect of borrowing constraint shocks on the labour market. In addition, this paper shows that, under labour and financial frictions, a binding EBITDA borrowing constraint raises the capital-labour ratio in the steady state and can raise the fundamental surplus via a higher marginal product of labour.

Finally, this paper is also related to the literature on the fundamental surplus, which Ljungqvist and Sargent (2017) show that the devices the literature (Shimer, 2005, 2004; Hall, 2005; Hagedorn and Manovskii, 2008) uses to answer the Shimer puzzle all work in the same way: they lower fundamental surplus. This paper contributes to this literature by showing that with EBITDA borrowing constraints and costly dividend adjustment, a low fundamental surplus also amplifies the response of financial and labour market following borrowing constraint shocks, thus it does not only amplify the labour market's response to productivity shocks.

The rest of the paper is structured as follows. Section 2 presents the empirical motivation, and Section 3 presents the model and analytically demonstrates how the interactions between financial and labour frictions establish a labour-financial market transmission channel. Section 4 analyses the role that the financial and labour frictions have on the economy's steady state, while Section 5 analyses their effects on the economy's dynamics. Section 6 analyses how financial frictions affect the fundamental surplus, and how a low fundamental surplus can amplify the labour-financial market transmission channel. Section 7 concludes.

2 Empirical Motivation

This section presents the empirical motivation. The percentage change in labour market tightness ($\Delta \log \theta_t$) and the net percentage of banks tightening commercial and industrial lending standards ($\Delta \ell_t$), two measures of conditions in the labour and financial markets, move inversely to one another, which is visualised in Figure 1. When the labour market tightens, banks loosen their lending standards, and vice versa. In addition, as Table 1 shows, the two measures are strongly contemporaneously correlated. Taken together, these two indicators suggest an economic link between labour and financial markets, through which shocks are transmitted rapidly. Prior studies have documented spillovers between the two markets. Chodorow-Reich (2014), for example, uses the financial health of firms' lenders at the time of Lehman Brothers' collapse as an exogenous shock to credit supply, and attributes between one-third and one-half of the subsequent employment decline at small and medium-sized firms to the withdrawal of credit. Bell and Machin (2018), using an event study, show that an unanticipated minimum wage hike announcement in the UK on the 8th July, 2015, significantly reduced stock market

values for low-wage firms. The evidence, therefore, suggests that labour and financial markets are closely connected and cannot be analysed in isolation.

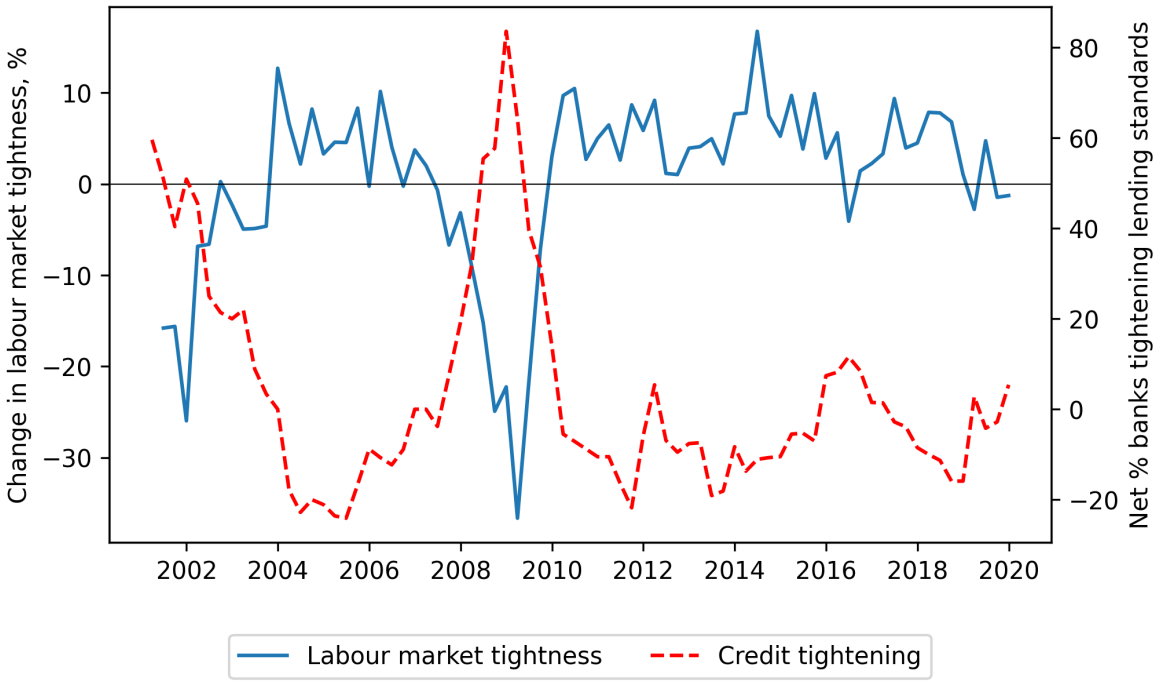


Figure 1: Percentage change in labour market tightness against net percentage credit tightening, 2001Q2 to 2019Q4. Labour market tightness is computed as JOLTS job openings per CPS unemployed worker, while credit tightening is plotted as the net percentage of banks tightening C&I lending standards over the quarter. Source: (U.S. Bureau of Labor Statistics, 2026a,b; Board of Governors of the Federal Reserve System, 2026).

Despite the close connections between the two markets, the workhorse labour-market model, the Diamond-Mortensen-Pissarides (DMP) model, oversimplifies the financial market. In the model, hiring is treated as an inter-temporal investment. The quantity of hiring, and, in large part, the firm's value of a match, therefore, depends on expected future marginal benefits of workers, which, if large enough, let firms maintain matches under negative cash flow, financing them implicitly by issuing equity costlessly. The drawback of this approach to the labour market is that, first, it fails to capture the costs of issuing equity, which will affect the firm's cash flow and almost certainly its labour

Table 1: Correlation between the change in labour market tightness, $\Delta \log \theta_t$, and the net percentage of banks tightening C&I lending standards i quarters earlier, $\Delta \ell_{t-i}$, 2001Q2 to 2019Q4. Source: (U.S. Bureau of Labor Statistics, 2026a,b; Board of Governors of the Federal Reserve System, 2026).

i	-4	-3	-2	-1	0	1	2	3	4
$\text{Corr}(\Delta \log \theta_t, \Delta \ell_{t-i})$	-0.380	-0.550	-0.710	-0.829	-0.859	-0.827	-0.718	-0.545	-0.325

demand and the wages it pays. Second, the model ignores firms selling debt, another oversimplification that causes it to overlook how frictions associated with borrowing can affect hiring and wages. Turning to financial market models, particularly those that model returns, like the CAPM, they also tend to overlook the role of search-frictional labour markets. Evidence suggests that labour search frictions strongly determine the cross-section of equity returns (Kuehn, Simutin and Wang, 2017). Intuitively, in frictional labour markets, worker flows are costly to firms and ultimately affect their cash flows.

3 The Model

There is a continuum of firms and households, each of unit mass. Firms are perfectly competitive, identical, and large: they use capital and labour as production inputs, invest in and own their own capital, post costly vacancies in a search-frictional labour market, and sell corporate debt. Households consist of a continuum of workers, also of unit mass. Employed workers earn a Nash-bargained wage, and unemployed workers earn a constant level of home production income. The household pools income, perfectly shares unemployment risk from the search-frictional labour market, and chooses consumption and corporate bond purchases.

3.1 Firms

The firm j, at time t, uses capital k_{t-1}^j and labour, n_t^j, to produce output y_t^j, according to,

$$y_t^j = z_t F(k_{t-1}^j, n_t^j) = z_t (k_{t-1}^j)^\alpha (n_t^j)^{1-\alpha} \tag{1}$$

where z_t is a stochastic economy-wide neutral productivity and F is the Cobb-Douglas production function. Capital depreciates at rate δ, so investment is $k_t^j - (1-\delta)k_{t-1}^j$.

The firm posts vacancies v_t^j at constant cost $\gamma > 0$ and loses a constant fraction $s \in (0,1)$ of workers at a constant separation rate. Let q_t denote the probability that a vacancy is filled, so the law of motion of labour is

$$n_{t+1}^j = (1-s)n_t^j + q_t v_t^j \tag{2}$$

Aggregate matches follow the Cobb-Douglas matching function,

$$H_t = \Upsilon V_t^{1-\eta} U_t^\eta \tag{3}$$

where V_t is aggregate vacancies, U_t aggregate unemployment, Υ is the matching efficiency and η is the elasticity of matches with respect to unemployment. The labour market's tightness, vacancy filling rate and the job finding rate therefore are

$$\theta_t = \frac{V_t}{U_t}, \qquad q_t = \frac{H_t}{V_t} = \Upsilon \theta^{-\eta}, \qquad f_t = \frac{H_t}{U_t} = \Upsilon \theta^{1-\eta} \tag{4}$$

with $q_t, f_t \in (0,1)$.

The firm sells risk-free one-period bonds x_t^j to households and receives $1/\tilde{R}_t$ per bond

to fund investment and smooth dividends. Following Jermann and Quadrini (2012) the firm receives a tax-benefit on the interest they pay on debt. $\tilde{R}_t$ is the effective gross interest rate that the firm pays on its debt, and the tax benefit is $\tau > 0$, such that,

$$\tilde{R}_t = 1 + (R_t - 1)(1 - \tau)$$

where $R_t = 1 + r_t$ is the gross interest rate households receive. Since,

$$\Delta_t = \frac{x_t}{\tilde{R}_t} - \frac{x_t}{R_t} > 0$$

the government transfers, $T_t = -\Delta_t$, to households to pay for the tax benefit.

The firm's borrowing is constrained by a multiple its EBITDA,

$$\frac{x_t^j}{\tilde{R}_t} \leq \xi_t \left(y_t^j - w_t^j n_t^j - \gamma v_t^j \right) \tag{5}$$

where w_t^j is the wage. The term in brackets $(y_t^j - w_t^j n_t^j - \gamma v_t^j)$ is the accounting definition of the firm's EBITDA. $\xi_t > 0$ is the EBITDA multiple, which can be interpreted as the number of periods of current earnings a lender will advance. In this model, borrowing constraint shock is a shock to ξ. A tightening shock, for instance, is one that reduces ξ. From (5), a growing labour force increases the firm's borrowing capacity if and only if the marginal product of labour exceeds the marginal cost. An important characteristic of (5) is that it makes the firm's borrowing capacity flow-based rather than stock-based, like a Kytatki-Moore (KM) borrowing constraint, which will be used as a comparison,

$$\frac{x_t^j}{\tilde{R}_t} \leq \xi_t^{KM} (1 - \delta) k_t^j \tag{6}$$

in which the firm's borrowing is collaterised against its undepreciated capital in the following period. ξ_t^{KM} is the fraction of the undepreciated capital that is used as collateral[1]. A firm that is borrowing constrained by (5) can quickly adjust its borrowing capacity by adjusting the vacancies it posts. A firm borrowing constrained by (6), conversely, will see its borrowing capacity adjust more slowly following shocks.

Households hold constant and equal shares in every firm, receive profits as dividends, and instruct firms to maximise the present value of the lifetime stream of profits. Following Jermann and Quadrini (2012), firm j pays dividends d_t^j and bears a quadratic cost when it deviates from the long-run target ϕ. The cost of paying dividends follows,

$$\psi(d_t^j) = d_t^j + \kappa \left(d_t^j - \phi\right)^2 \tag{7}$$

where $\kappa \geq 0$ is the adjustment cost parameter, which can be interpreted as the cost of substituting between raising finance with debt and reducing dividends. Alternatively, κ can capture managers' preference for smoothing dividends, which Jermann and Quadrini indicate has been confirmed in the literature and can arise from agency problems. In this case, the larger the κ, the stronger the managers' preference for smoothing dividends. κ has no effect on a steady state when ϕ is set equal to the steady state dividends.

The firm's flow budget, therefore, is,

$$y_t^j - w_t^j n_t^j - \gamma v_t^j + \frac{x_t^j}{\tilde{R}_t} = x_{t-1}^j + k_t^j - (1-\delta)k_{t-1}^j + \psi(d_t^j) \tag{8}$$

Using (7) and (8), it can be shown that when the borrowing constraint binds, if the firm's expenditures exceed its income, it must reduce its dividends. Suppose the required

[1]In steady state, ξ^{KM} is set to one.

reduction in dividends is from ϕ to $\phi - \Delta_d$, where $\Delta_d > 0$; then, given the adjustment cost, the resources that become available to the firm following the reduction are

$$\Delta_d - \kappa \Delta_d^2$$

As κ increases, the maximum resources the firm can obtain through dividend reductions decreases, and the firm must cut more dividends to achieve the same desired increase in available resources. If $\kappa = 0$, then the firm can obtain any desired increase in resources by cutting dividends by the same amount, and even raise equity via negative dividends.

3.2 The Firm's Problem, Recursively

The aggregate state is,

$$S \equiv (z, \xi, N, K, X)$$

where N is aggregate employment, K is aggregate capital, and X is aggregate debt, which evolves according to $S' = \Gamma(S, z', \xi')$ The firms invididual state is,

$$\omega \equiv (n, k, x)$$

The firm's optimisation problem is,

$$J(\omega; S) = \max_{d,v,k',x'} \{d + \mathbb{E}\left[m' J(\omega'; S')\right]\} \tag{9}$$

where m' is the owner's stochastic discount factor, subject to the flow budget

$$y - wn - \gamma v + \frac{x'}{\tilde{R}} = x + k' - (1 - \delta)k + \psi(d)$$

the employment law of motion (2), and the EBITDA borrowing constraint

$$\frac{x'}{\tilde{R}} \leq \xi \left(y - wn - \gamma v \right)$$

Remark 1. With $\tau > 0$, the firm will find it profitable to finance itself with debt over equity; thus, the borrowing constraint will bind. Combining the binding borrowing constraint with the budget yields,

$$1 + \xi = \frac{x + k' - (1 - \delta)k + \psi(d)}{y - wn - \gamma v} > 1 \tag{10}$$

(10) shows that, when the borrowing constraint binds, ξ determines the percentage difference between firms' debt repayments, investment, and dividends and their EBITDA. Using these constraints, the Bellman may be written in the form,

$$J(\omega; S) = \max_{d,v,k',x'} \Bigg\{ d + \mathbb{E}\left[m' J(\omega'; S') \right] + \mu \left[y - wn - \gamma v + \frac{x'}{\tilde{R}} - x - k' + (1 - \delta)k - \psi(d) \right] + \Lambda \left[\xi(y - wn - \gamma v) - \frac{x'}{\tilde{R}} \right] \Bigg\}$$

Here, μ can be interpreted as the value of an additional unit of internal funds available to the firm[2], and Λ is the value of loosening the borrowing constraint. From the two constraints, increases in output raise the firm's value function through the budget and borrowing constraints in the current period. Similarly, labour and hiring costs lower the firm's value function through both constraints in the current period. Variables not in EBITDA, such as dividends, debt and capital investment, affect the firm's value differently. Dividends, for instance, only affect the firm's value through the current marginal value of internal funds. In the current period, capital investment reduces the firm's value by tightening the budget constraint, but in the following period, it raises the firm's value through more internal resources and a looser borrowing constraint. Debt

[2]Alternatively, one can interpret this as how hard the firm's budget constraint binds

sales, on the other hand, raise the firm's value by providing more internal funds today; this is offset by a tighter borrowing constraint today and fewer internal funds available in the future, which reduce the firm's value.

3.3 The Labour–Finance Transmission Mechanism Through the Firm's Optimality Conditions

The section below examines the labour-financial market link in the EBITDA economy through the firm's optimality conditions, and compares them to the optimality conditions in the KM economy. For the full derivations of the following optimality conditions in the EBITDA economy, see Appendix A. For the KM economy's optimality conditions, see Appendix F.

Dividends. The marginal benefit of dividends to the firm owner is unity, while the marginal cost consists of the costs of paying the dividend, $\psi'(d)$, priced at μ. The optimality condition for dividends is, therefore,

$$1 = \mu\psi'(d) \tag{11}$$

where $\psi'(d) = 1 + 2\kappa(d - \phi)$. In the following analysis, let $\mu \equiv 1 + \hat{\mu}$, which means (11) becomes,

$$1 + \hat{\mu} = \frac{1}{1 + 2\kappa(d - \phi)}$$

after rearranging,

$$\hat{\mu} = \frac{-2\kappa(d - \phi)}{1 + 2\kappa(d - \phi)} \tag{12}$$

$\hat{\mu}$ shows the budget constraint's bite compared to the case where $\kappa = 0$. If dividends are at their target, $d = \phi$, then the budget constraint's bite is the same as in the $\kappa = 0$ case.

When dividends are below target, $d < \phi$, then $\hat{\mu} > 0$, as the firm is in financial distress. In this case, it values an additional unit of internal funds more highly than in the $\kappa = 0$ case, because the budget constraint bites harder. Conversely, if dividends are above target, $d > \phi$, then $\hat{\mu} < 0$, and the budget constraint bites less than in the $\kappa = 0$ case.

Vacancies. The cost of posting a vacancy for the firm includes the unit cost γ, which reduces the firm's value by $\gamma(1 + \hat{\mu})$ units, and the $\xi\gamma$ loss of borrowing capacity, valued at $\xi\gamma\Lambda$, resulting from a tighter borrowing constraint. Consequently, the marginal cost of posting vacancies is $\gamma(1 + \hat{\mu} + \xi\Lambda)$, while the marginal benefit is the discounted value of an additional worker, given the probability q that the vacancy is filled. At the optimum, the following condition must hold,

$$q\mathbb{E}\left[m' J_n(\omega'; S')\right] = \gamma(1 + \hat{\mu} + \xi\Lambda) \tag{13}$$

This condition is an extended version of the standard DMP free-entry condition. If dividends are below target or the borrowing constraint bites harder, the marginal cost of hiring rises, and the firm posts fewer vacancies. When dividends are at their target, or $\kappa = 0$, and the firm has no tax advantage on debt (i.e., $\tau = 0$ and $\Lambda = 0$), the borrowing constraint does not bind, and the condition reduces to the standard DMP free-entry condition.

Compare (13) to the vacancy optimality condition if the firm's borrowing constraint is (6),

$$q\mathbb{E}\left[m' J_n(\omega'; S')\right] = \gamma(1 + \hat{\mu})$$

Under a KM borrowing constraint, the borrowing constraint's tightness has no direct effect on the marginal cost of posting vacancies. The way in which the KM borrowing

constraint affects the firm's hiring is via the changes in the future expected marginal product of capital on the future expected marginal product of labour. Alternatively, if changes in the firm's capital investment change $\hat{\mu}$.

Workers. An additional worker raises the firm's output by the marginal product of labour, less the wage, which in the current period generates

$$(1 + \hat{\mu} + \xi\Lambda)\left[zF_n(k, n) - w\right]$$

of value to the firm. This means that if the firm has low dividends, or the borrowing constraint bites harder, the firm values the net income generated by the worker today more. The value of an additional worker can be obtained from the envelope condition with respect to n, and thus obeys,

$$J_n(\omega; S) = (1 + \hat{\mu} + \xi\Lambda)\left[zF_n(k, n) - w\right] + (1 - s)\mathbb{E}\left[m' J_n(\omega'; S')\right] \tag{14}$$

This is analogous to the value of a filled vacancy in the DMP model.

If the firm were to be KM borrowing constrained, the envelope condition would be,

$$J_n(\omega; S) = (1 + \hat{\mu})\left[zF_n(k, n) - w\right] + (1 - s)\mathbb{E}\left[m' J_n(\omega'; S')\right]$$

As seen before, the KM borrowing constraint's tightness has no direct effect on the value of an additional worker for the firm.

Capital. The marginal cost of purchasing an additional unit of capital is $1 + \hat{\mu}$. The marginal benefit of capital investment is the additional units of future resources it generates, its marginal product, plus its undepreciated units, valued at $1 + \hat{\mu}'$. It also

raises the firm's borrowing capacity by raising its output, which the firm values at Λ'. At the optimum,

$$\mathbb{E}\Big[m'\Big\{(1+\hat{\mu}')\Big[z'F_k(k',n')+(1-\delta)\Big]+\xi'\Lambda'z'F_k(k',n')\Big\}\Big]=1+\hat{\mu} \tag{15}$$

which is the firm's capital Euler equation. Financial frictions change capital demand. For example, if dividends are expected to fall or the firm's borrowing capacity is expected to diminish, the marginal benefit of capital investment rises, since its return can help smooth dividends and support the firm's borrowing capacity.

Remark 2. When dividends are at their target, and the borrowing constraint bites, $\Lambda > 0$, the capital-labour ratio will be larger than when $\hat{\mu} = \Lambda = 0$. Using the Cobb-Douglas production function, and letting ρ be the discount rate, the capital-labour ratio in the steady state is,

$$\frac{k}{n}=\left[\frac{\alpha z(1+\xi\frac{\Lambda}{1+\hat{\mu}})}{\rho+\delta}\right]^{\frac{1}{1-\alpha}} \tag{16}$$

When the borrowing constraint binds, and $\hat{\mu} = 0$, capital investment raises the firm's borrowing capacity and the firm's desired steady state capital-labour ratio. When the borrowing constraint does not bind, and $\hat{\mu} = 0$, then (16) reduces to the well-known neoclassical growth model capital-labour ratio.

(16) also shows the capital-labour ratios in the hypothetical cases where dividends are not at their target in the steady state. For instance, if dividends are below target in the steady state, $\hat{\mu} > 0$, then the capital-labour ratio is smaller because firms wish to recover earnings to raise dividends to their target. Alternatively, if dividends are above target in the steady state, $\hat{\mu} < 0$, the capital-labour ratio is larger as the firm wishes to increase its total production costs to bring dividends down to target.

If the firm were to be constrained by the KM borrowing constraint, capital's optimality

condition would be instead,

$$\mathbb{E}\left[m'\Big\{(1+\hat{\mu}')\left[z'F_k(k',n')+1-\delta\right]\Big\}\right]+\Lambda\xi^{KM}(1-\delta)=1+\hat{\mu}$$

In contrast to (15), the marginal benefit of capital arises in the current period from loosening the borrowing constraint today and from the additional expected future funds generated by the newly installed capital in the following period. As a result, the firm's investments are less forward-looking than under EBITDA borrowing constraints. For example, if the firm were to expect a larger Λ in the future, it would increase its investments in anticipation. In the KM economy, the expected future tightness of the borrowing constraint does not change the firm's behaviour.

Debt. A unit of borrowing delivers $1/\tilde{R}$ of resources to the budget, adding $(1+\hat{\mu})/\tilde{R}$ units of value, and consumes $1/\tilde{R}$ of borrowing capacity, worth $\Lambda/\tilde{R}$ units of value. The optimality condition is,

$$\frac{\Lambda}{\tilde{R}}+\mathbb{E}\left[m'(1+\hat{\mu}')\right]=\frac{1+\hat{\mu}}{\tilde{R}} \tag{17}$$

which is the same as in the KM borrowing constraint case. (17) is the firm's debt Euler equation. The tax advantage associated with debt financing ensures that $\tilde{R}<R$ and $\Lambda>0$. In the steady state, where $m'=1/R$ and $\hat{\mu}'=\hat{\mu}=0$, then,

$$\Lambda=1-\frac{\tilde{R}}{R}$$

If $\tau=0$, then $\tilde{R}=R$, and $\Lambda=0$, in which case the firm's equity-debt choice would be indeterminant.

3.4 Households

A continuum of identical households, each of unit mass, holds a continuum of workers, also of unit mass, of which n are employed and $1 - n$ are unemployed. Employed workers earn a Nash-bargained wage w, and unemployed workers receive a constant home-production income b. Workers separate at rate s and find jobs at rate f, but they pool their income inside the household, so unemployment risk is perfectly shared. Households own constant and equal shares in every firm, earning the dividend d, and buy a' units of the bonds firms sell at the price $1/R$. The government funds the firms' interest deduction with a lump-sum tax levied uniformly on households,

$$T = \frac{X'}{R} - \frac{X'}{\tilde{R}},$$

where X' is the aggregate quantity of bonds traded.

Households have CRRA preferences over consumption, c,

$$u(c) = \frac{c^{1-\sigma}}{1-\sigma}$$

where σ is the CRRA parameter. $W(a, n; S)$ is the household's value function, hence its optimisation problem is,

$$W(a, n; S) = \max_{c,a'} \left\{ u(c) + \beta \mathbb{E} \left[W(a', n'; S') \right] \right\}, \tag{18}$$

subject to the budget

$$c + \frac{a'}{R} = wn + (1 - n)b + a + d + T \tag{19}$$

and the law of motion of labour,

$$n' = (1-s)n + f(1-n) \tag{20}$$

Bond holdings satisfy the Euler equation,

$$u'(c) = \beta R \, \mathbb{E}\left[u'(c')\right] \tag{21}$$

which prices the one-period risk-free bond and makes the gross real rate R move with the consumption path. Using this Euler equation, the firm's discount factor is, $m' = \beta u'(c')/u'(c)$.

3.5 Nash Wages

The wage is determined through Nash bargaining between the worker and the firm,

$$w = \arg\max_{w} \left\{ \left(W_n(a,n;S)\right)^{\Psi} J_n(\omega;S)^{1-\Psi} \right\} \tag{22}$$

where $W_n(a,n) = \partial W(a,n)/\partial n$ is the household's marginal value of employment and Ψ is the worker's bargaining power. The resulting frictional Nash wage is[3]

$$w = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} \left[1 - \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right] \right] + \gamma\theta\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right] \right] + (1-\Psi)b \tag{23}$$

where, in the steady state when $\hat{\mu} = \hat{\mu}'$, $\xi' = \xi$ and $\Lambda = \Lambda' > 0$, or in the absence of frictions, which is when $\hat{\mu} = \Lambda = 0$, the wage simplifies to,

$$w^f = \Psi \left[zF_n(k,n) + \theta\gamma \right] + (1-\Psi)b$$

[3] See Appendix B for the derivation.

where w^f stands for the frictionless wage. Unlike Monacelli, Quadrini and Trigari (2023), debt does not enter the wage because it does not affect the value of an additional worker to the firm or the value of an additional employed worker to the household, and therefore is not part of the joint surplus. Financial frictions, $\hat{\mu}$ and Λ, enter the wage, however. To see the effect of the financial frictions on the wage, subtract the frictionless wage from (23), and letting $\Omega \equiv \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right]$, which yields,

$$w - w^f = \Psi \frac{\gamma}{q} \left[(1-s) - f\right](1-\Omega) \gtreqless 0$$

In the absence of financial frictions, $\Omega = 1$, and then $w - w^f = 0$. If $(1-s) > f$, then a match is more likely to remain in each period than an unemployed worker is likely to find a match. If $\Omega < 1$, then the frictional wage rises over the frictionless wage. In this case, future funds and borrowing capacity become relatively more valuable, and the firm begins valuing the incumbent workforce more because they generate output and support its borrowing capacity without the firm having to undertake costly hiring. Conversely, if $\Omega > 1$, then the frictional wage falls below the frictionless wage.

Turning to the case where $(1-s) < f$, the effects reverse. New matches are easier to create, so changes in financial conditions affect the value the firm assigns to its incumbent workforce in the opposite way.

In the case of the KM borrowing constraint, the wage is

$$w = \Psi \left[z F_n(k,n) + \frac{(1-s)\gamma}{q} \left[1 - \mathbb{E}\left[\frac{1+\hat{\mu}}{1+\hat{\mu}'}\right]\right] + \gamma\theta\mathbb{E}\left[\frac{1+\hat{\mu}}{1+\hat{\mu}'}\right]\right] + (1-\Psi)b$$

As expected, the wage in the KM economy is only indirectly affected by the borrowing constraint, via the changes in the value of internal funds, $\hat{\mu}$, or changes to the marginal product of labour.

Definition 3.1 (Recursive Competitive Equilibrium)**.** A recursive competitive equilibrium in the EBITDA economy consists of the set of household's value function $W(a, n; S)$ and its policy functions $c(a, n; S)$, $a'(a, n; S)$, the firm's value function $J(\omega; S)$ and its policy functions $d(\omega; S)$, $v(\omega; S)$, $k'(\omega; S)$, $x'(\omega; S)$, aggregate prices $R(S)$, $w(S)$, and $m(S, S')$, labour market tightness $\theta(S)$ and the law of motion of aggregate states $S' = \Gamma(S)$, such that,

1. **Firm Optimisation.** Given $R(S)$, $w(S)$, and $\theta(S)$, and the aggregate law of motion Γ, the firm's policies are optimal and $J(\omega; S)$ solves the firm's Bellman equation.

2. **Household Optimisation** Given $R(S)$, $w(S)$, and $\theta(S)$, the household's policies are optimal and $W(a, n; S)$ solves the household's Bellman equation.

3. **Wage Determination.** The wage $w(S)$ solves the Nash bargaining problem (22).

4. **Aggregation and Matching.** By symmetry, $n = N$, $k = K$ and $x = X$. This means in the aggregate, $Y(S) = zF(K, N)$, $D(S) = d(N, K, X; S)$, $V(S) = v(N, K, X; S)$, $K'(S) = k'(N, K, X; S)$, $U(S) = 1 - N$, $X'(S) = x'(N, K, X; S)$. Where $C(S)$ is aggregate consumption, the stochastic discount factor for firms is $m(S, S') = \beta u'(C(S'))/u'(C(S))$. Matches satsify $H(S) = \Upsilon V(S)^{1-\eta} U(S)^{\eta}$, thus $\theta(S) = V(S)/U(S)$, $q(S) = \Upsilon\theta(S)^{-\eta}$, and $f(S) = \Upsilon\theta(S)^{1-\eta}$. Labour evolves according to $N'(S) = (1 - s)N + q(S)V(S)$.

5. **Taxes and Market Clearing.** Households purchase firms' debt, thus, $a'(X, N; S) = X'(S)$ is the bond market clearing. This means $C(S) = c(X, N; S)$. Taxes satisfy,

$$T(S) = \frac{X'(S)}{R(S)} - \frac{X'(S)}{\tilde{R}(S)}$$

and the goods market clears according to,

$$Y(S) + (1 - N)b - \gamma V(S) = C(S) + K'(S) - (1 - \delta)K + \kappa(D(S) - \phi)^2$$

6. **Aggregate Law of Motion.** The aggregate law of motion $S' = \Gamma(S, z', \xi')$ is consistent with agents' decisions and z and ξ's stochastic processes.

4 Labour-Financial Market Link in the Steady State

This section analyses the role that the interactions between the financial frictions, specifically the borrowing constraints, and labour market frictions have on an economy's steady state.

4.1 Parametrising the EBITDA economy's steady state

The vacancy cost is calibrated to 0.11 and matching efficiency to 0.31, and the separation rate is s is set to 0.02, targeting a steady state unemployment rate of $U = 6\%$, and a steady state labour market tightness of $\theta = 1$. Home production income is set to $b = 0.7$, and δ is set to 0.107 to ensure the dividend steady state remains non-negative while bringing the steady state marginal product of labour close to unity.

β is set to 0.985 and α is set to 0.36 following Jermann and Quadrini (2012). Following Fujita and Ramey (2012) the matching elasticity is set to $\eta = 0.7$ and a worker bargaining weight is $\Psi = 0.7$. The constant risk aversion parameter σ is set to 2 and the steady state productivity z is set to 1. Following evidence from Lian and Ma (2021), the steady state EBITDA multiple ξ is set to 9.12, meaning that, with each period representing a quarter, firms, in the steady state, can borrow up to 2.3 times their annual earnings.

Turning to the tax τ, rounding up the top statutory rate a US corporation faces once state taxes are added to the federal rate, which was 39.1% in 2012 (Congressional Budget Office, 2017), it is set to 0.4. This value is higher than Jermann and Quadrini's τ of 0.35 because the authors consider solely federal taxes. The dividend target ϕ is set to the dividend steady state.

4.2 Parametrising the Alternative Economies

The EBITDA economy's steady state is compared to two alternatives in the following analysis. The first is an otherwise identical frictionless baseline where $\tau = 0$ and firms face no borrowing constraint. In this economy, similar to the RBC model, firms borrow to invest in new capital. In this frictionless baseline, there are still DMP labour frictions because removing these would change the interpretation of labour to hours worked, making the model incomparable to the KM and EBITDA economies. The second alternative is a version where the borrowing constraint is asset-based and follows (6). The parameters γ and Υ are calibrated targeting the same unemployment rate and labour market tightness, set ϕ to the steady state level of dividends, and keep the remaining parameters at the same values as in the EBITDA economy. $\xi^{KM} = 1$ in the steady state.

4.3 Comparing the Steady States

Table 2 compares the steady states of the three economies.

Table 2: Steady states across the three economies, 2 d.p.

		Frictionless baseline	EBITDA Economy	KM Economy
Y	Output	1.73	1.78	1.77
zF_n	Marginal product of labour	1.18	1.21	1.21
w	Wage	1.16	1.19	1.18
U	Unemployment	0.06	0.06	0.06
θ	Tightness	1.00	1.00	1.00
Λ	Collateral multiplier	–	0.01	0.01
μ	Firm budget multiplier	–	1.00	1.00
X	Debt	5.08	5.97	4.92
K	Capital	5.08	5.52	5.46
d	Steady State Dividends	0.01	0.00	0.02

The EBITDA borrowing constraint affects the steady state through the capital-labour ratio, as shown in (16). The binding borrowing constraint raises the marginal product of labour through a larger K/N; the EBITDA economy's marginal product of labour is 1.211, while the KM economy's is 1.206. This raises the EBITDA economy's output and Nash wages, but because the larger capital stock raises total production costs, its dividends fall to 0.004.

5 Labour-Financial Market Link in the Dynamics

This section analyses the labour-financial transmission channel in EBITDA, KM and baseline economies, via impulse response shocks to z, ξ and ξ^{KM}. The shock processes

in this experiment for z, ξ and ξ^{KM} are,

$$
\begin{aligned}
z_t &= (1-\rho_z)z_{ss} + \rho_z z_{t-1} + \varepsilon_t^z, \\
\xi_t &= (1-\rho_\xi)\xi_{ss} + \rho_\xi \xi_{t-1} + \varepsilon_t^\xi, \\
\xi_t^{KM} &= (1-\rho_{\xi^{KM}})\xi_{ss}^{KM} + \rho_{\xi^{KM}}\xi_{t-1}^{KM} + \varepsilon_t^{\xi^{KM}}.
\end{aligned}
$$

The persistence are set such that $\rho_\xi = \rho_z = \rho_{\xi^{KM}} = 0.95$, and $\sigma_z = \sigma_\xi = \sigma_{\xi^{KM}} = 0.01$. For the shocks, $\varepsilon_t^z \sim N(0, (z_{ss}\sigma_z)^2)$, $\varepsilon_t^\xi \sim N(0, (\xi_{ss}\sigma_\xi)^2)$ and $\varepsilon_t^{\xi^{KM}} \sim N(0, (\xi_{ss}^{KM}\sigma_{\xi^{KM}})^2)$.

To identify the effects of interactions between borrowing constraints and labour frictions on dynamics, for each shock, the responses of the three economies are shown under $\kappa = 0$. To show the effects of dividend adjustment costs, borrowing constraints, and labour frictions on dynamics, the responses of the three economies under $\kappa = 0.03$ are presented as well. Treating Λ as the counterpart to the credit tightening measure in Figure 1, this non-negative κ is chosen so that the model's implied ratio of the standard deviation of $\Delta \log \theta$ to the standard deviation of $\Delta \log \Lambda$ is aligned with the ratio computed from the data presented, which is 0.4. Table 3 shows the implied standard deviations from the EBITDA and KM economies across different κ values for the productivity and borrowing constraint shocks.

Table 3: Relative volatility across values of κ

		Shock	
Model	κ	Productivity	Borrowing Constraint
EBITDA	0.05	0.178	0.048
EBITDA	0.03	0.229	0.052
EBITDA	0.01	0.409	0.061
EBITDA	0.00	0.181	0.058
KM	0.05	0.279	0.007
KM	0.03	0.326	0.007
KM	0.01	0.246	0.007
KM	0.00	0.204	0.037

Note: The entries report the $\text{sd}(\Delta \log \theta)/\text{sd}(\Delta \log \Lambda)$ computed from the impulse responses.

The results show that $\kappa = 0.01$ brings the EBITDA economy closest to target, while $\kappa = 0.03$ does for the KM economy, each following a productivity shock. Under borrowing constraint shocks, the large response of Λ, which increases with κ, brings each model's implied volatility ratio below the target. The comparative analysis therefore uses $\kappa = 0.03$ as the shared value because, at this level, both economies underestimate the standard deviation ratio by the most similar amount. Nevertheless, this paper's analysis and conclusions do not depend on this particular choice of κ.

5.1 Productivity Shock

Figures 2 and 3 show the percentage deviations of EBITDA, KM, and baseline economies for the two κ values following a 1% productivity shock.

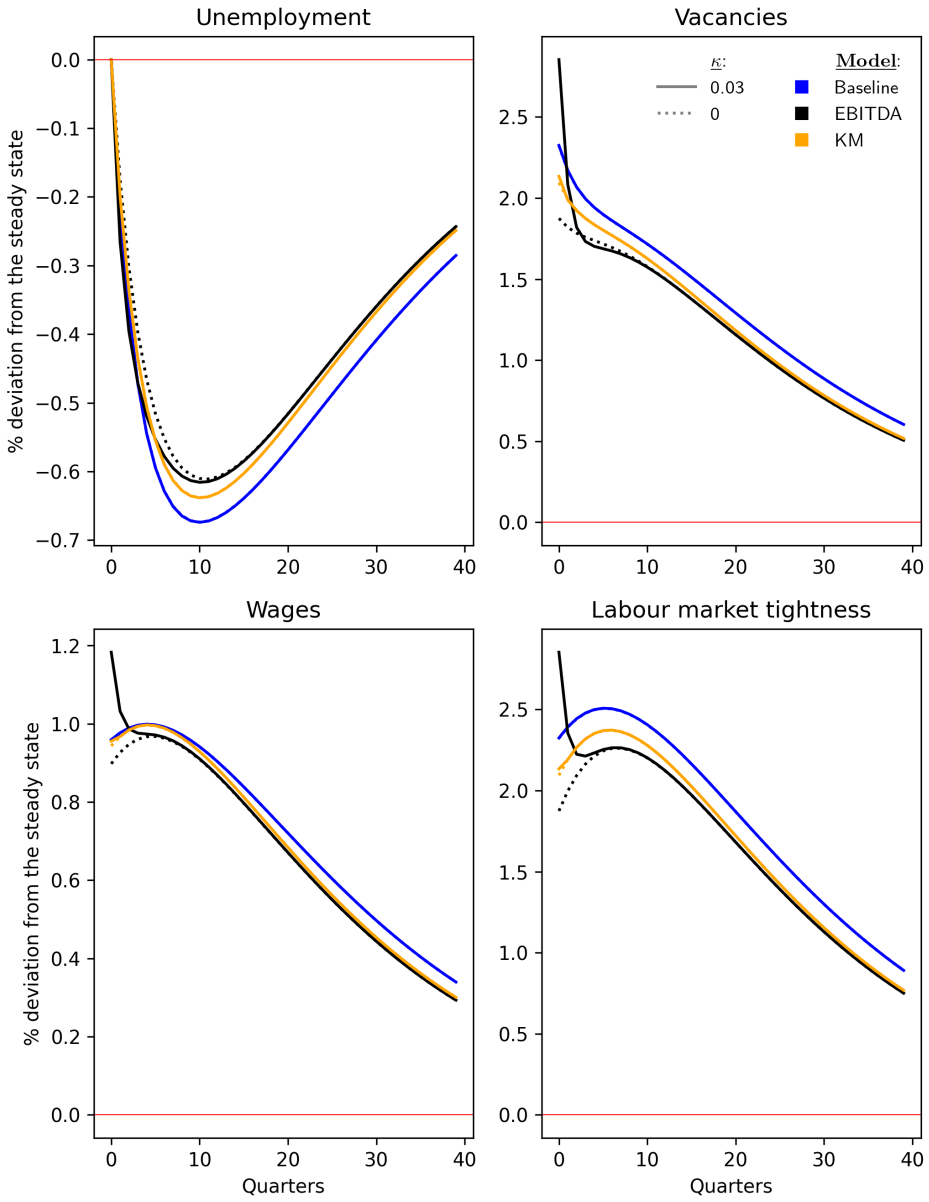


Figure 2: Percentage deviations from the steady state, 1% productivity shock.

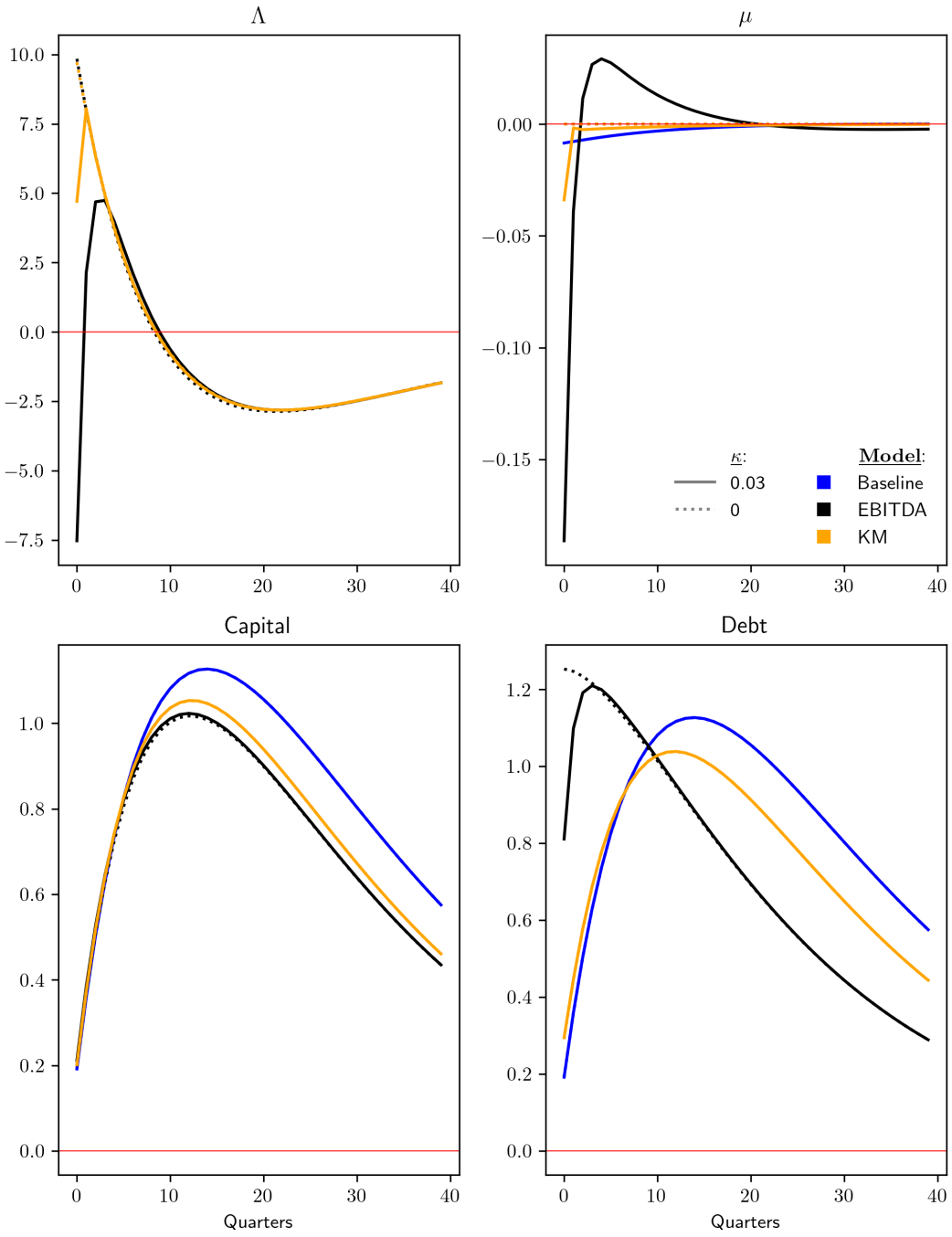


Figure 3: Percentage deviations from the steady state, 1% productivity shock.

The figures show two things. First, comparing the coloured lines shows that the type of borrowing constraint changes the economy's response, since the KM economy responses differ from the EBITDA economy responses. Second, comparing the dotted lines with the

solid lines of the same colour shows that dividend adjustment costs amplify the response primarily in the EBITDA-constrained economy. In both the KM and baseline economies, dividend adjustment costs do not alter the paths of unemployment, vacancies, wages and labour market tightness, nor the response of capital and debt. For the EBITDA economy, the solid black line deviates from the $\kappa = 0$ case. This result is consistent with the mechanisms identified from the firm's optimality conditions, because changes in $\hat{\mu}$ and Λ change the firm's value of labour and demand for hiring, which spill over into its cash flow, borrowing capacity and investment. The small effect that κ has on the KM economy may be explained by the slow adjustment of the firm's borrowing capacity. Intuitively, the slow-moving borrowing capacity cannot accommodate the firm's increased demand for hiring and capital, thus bringing the firm's response in the KM economy close to the $\kappa = 0$ case.

Turning now to the labour market, consider the baseline. Comparing the blue dotted to the blue solid lines, dividend adjustment costs alone cannot generate an amplification. Dividend adjustment costs only amplify the labour market's response in the EBITDA economy. At $\kappa = 0.03$, the EBITDA economy's labour market tightness and wage responses are between 30% and 50% larger than when $\kappa = 0$. After the shock hits, these responses are also 20% to 35% larger than in the baseline and KM economies. The amplification in the EBITDA economy can be explained by the marginal cost of vacancies, $\gamma(1 + \hat{\mu} + \xi\Lambda)$. The productivity shock increases the firm's resources, lowering $\hat{\mu}$, and expands the firm's borrowing capacity, reducing Λ. This mechanism raises vacancies in response to the shock. The same amplification does not occur in KM economy because changes in the firm's cash flow do not affect its borrowing capacity, and its slow-moving nature does not allow dividends to deviate as far from the target as in the EBITDA economy.

Turning to the financial market responses, the productivity shock immediately increases the EBITDA-constrained firm's borrowing capacity. Whether the firm has dividend adjustment costs or not influences how much the firm finances its expanded productive capacity with debt. For example, when $\kappa = 0$, after the shock the firm increases its dividend payments and relies more on borrowing to finance its vacancies and investments, so Λ increases. With dividend adjustment costs, the firm borrows less and finances its investments and hiring with internal funds. As a consequence, there is a lesser demand for debt, and a lower Λ upon the shock's impact, in comparison to the no dividend adjustment cost case.

The KM economy responds differently because the shock does not immediately increase the firm's borrowing capacity. Instead, over the shock, the borrowing capacity gradually expands as the firm accumulates capital. In this case, dividend smoothing has little effect on the firm's borrowing capacity, so κ has a weaker effect on $\hat{\mu}$ and Λ.

5.2 Borrowing Constraint Shock

Figures 4 and 5 show the effects of a -1% shock to ξ and ξ^{KM} on the EBITDA and KM economies, for the two different κ values as before. The baseline economy does not change following this shock because it has no borrowing constraint.

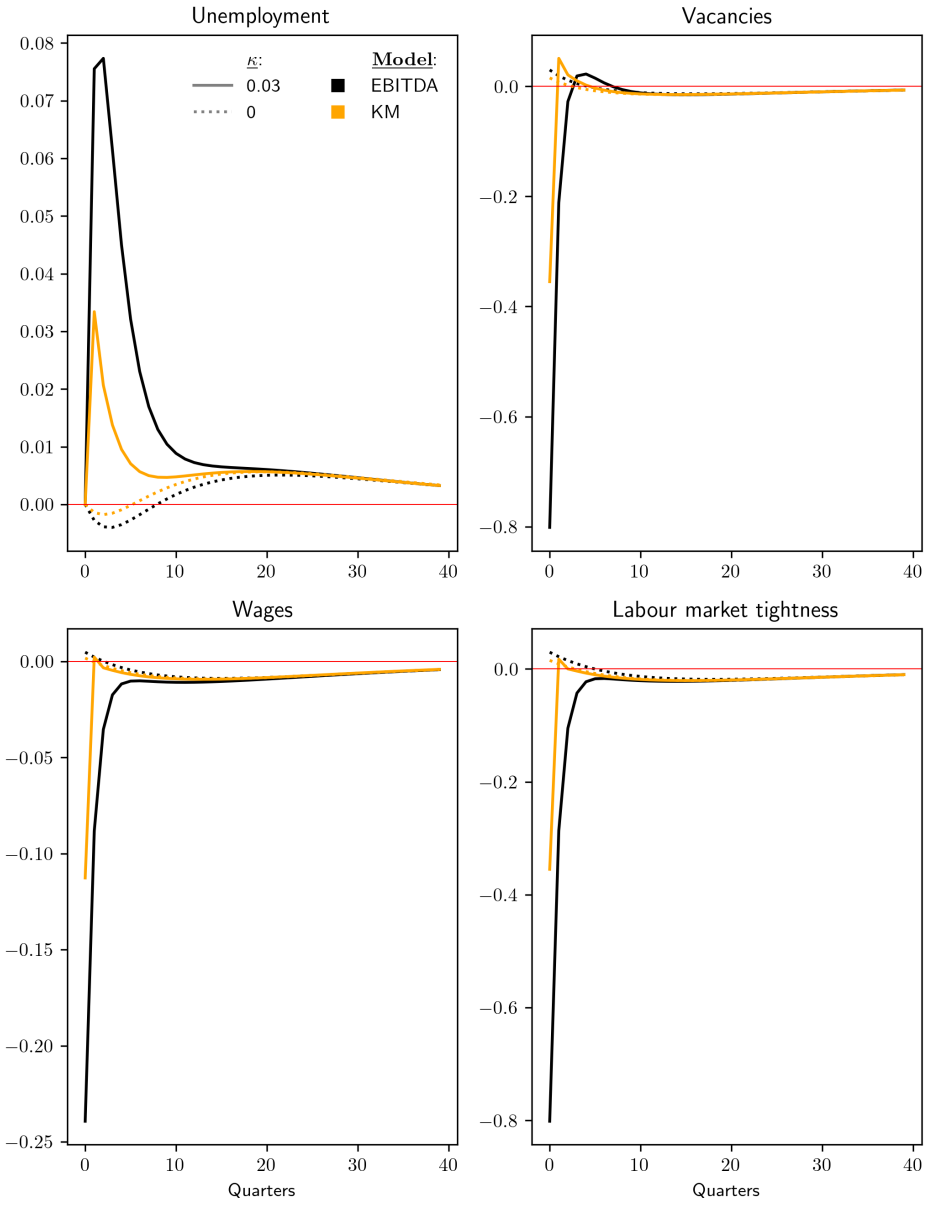


Figure 4: Percentage deviation from the steady state, −1% borrowing constraint shock.

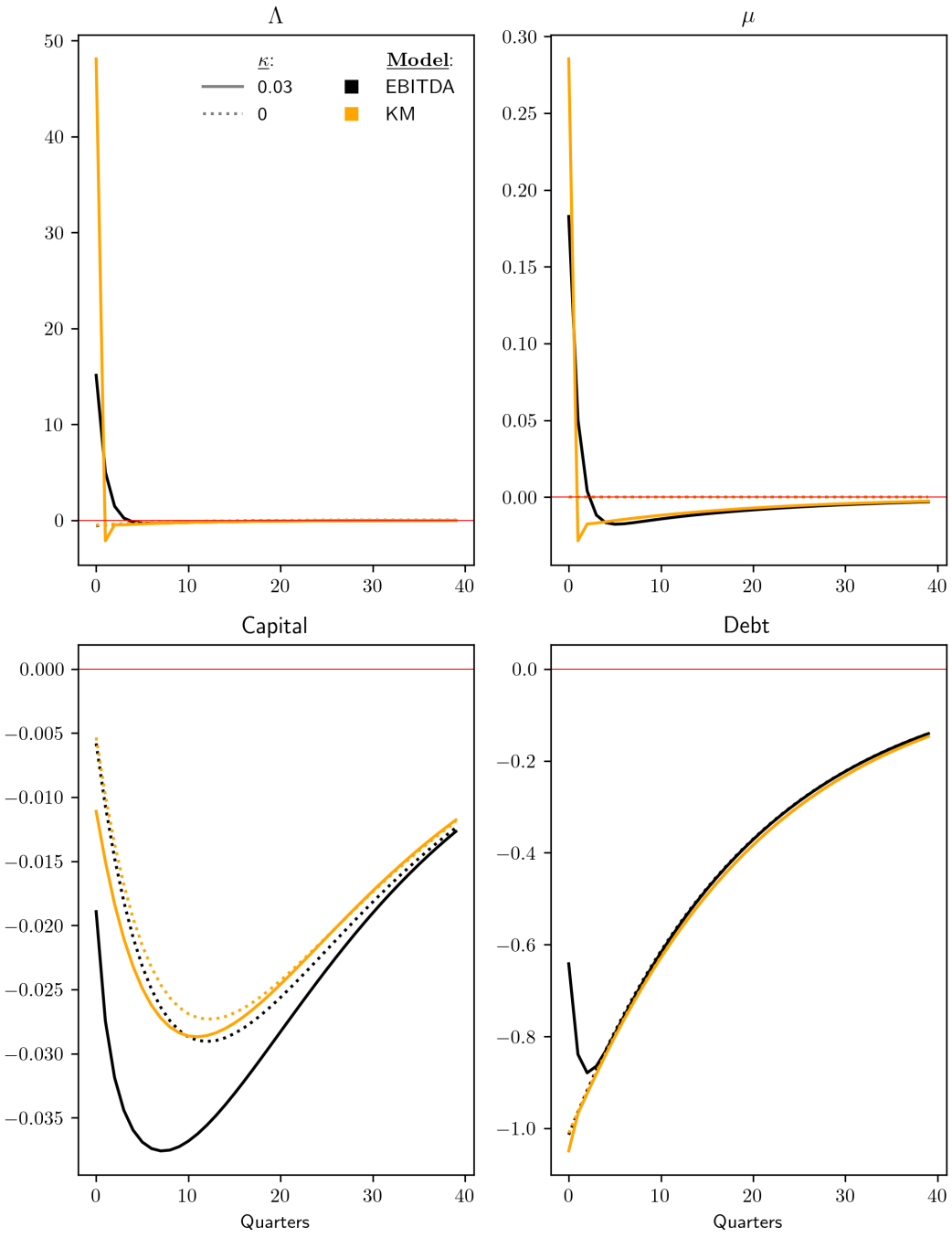


Figure 5: Percentage deviation from the steady state, −1% borrowing constraint shock.

The shock reaches the labour market only when there are dividend adjustment costs. Without dividend adjustment costs, the shock's effect on vacancies and wages is small, as seen in the small deviations of the dotted lines. With dividend adjustment costs, the

changes in both Λ and $\hat{\mu}$ begin to meaningfully affect the firm's hiring decisions. As with the productivity shock, the EBITDA and KM borrowing constraints transmit the shock differently: the former is based on a flow, while the latter is based on a stock.

With κ set to zero, the borrowing constraint shock does not transmit to the labour market, as vacancies and wages respond minimally to the shock. In addition, the borrowing constraint shock reduces the borrowing constraint's tightness. This is because with $\hat{\mu} = \hat{\mu}' = 0$, Λ evolves according to the path of the gross real interest rate,

$$\Lambda = 1 - \frac{1 + (R-1)(1-\tau)}{R}$$

where for $R > 0$, $\partial\Lambda/\partial R > 0$, which is intuitive: the larger the real interest rate that households receive on debt, the tighter the borrowing constraint. Since the gross rate evolves according to the Euler equation,

$$R = \frac{1}{\beta} \frac{u'(c)}{\mathbb{E}\left[u'(c')\right]}$$

the mechanism that reduces Λ under the borrowing-constraint shock is the expected future path of declining consumption, which increases demand for bonds as workers smooth consumption and reduces the real interest rate workers receive and firms pay, thereby reducing the bite of the firm's borrowing constraint. Once $\kappa > 0$, unemployment rises, and wages fall in both the EBITDA and KM economies following the shock. The mechanism is that the borrowing-constraint-tightening shock increases $\gamma(1 + \hat{\mu} + \xi\Lambda)$; thus, the cost of posting vacancies rises for the firm, and the firm cuts vacancies. The structure of the borrowing constraint, however, shapes the response: the KM economy's response in μ, Λ and debt is larger than under the EBITDA constraint. This is because, after the shock, firms in the KM economy must reduce investment and borrowing. Their

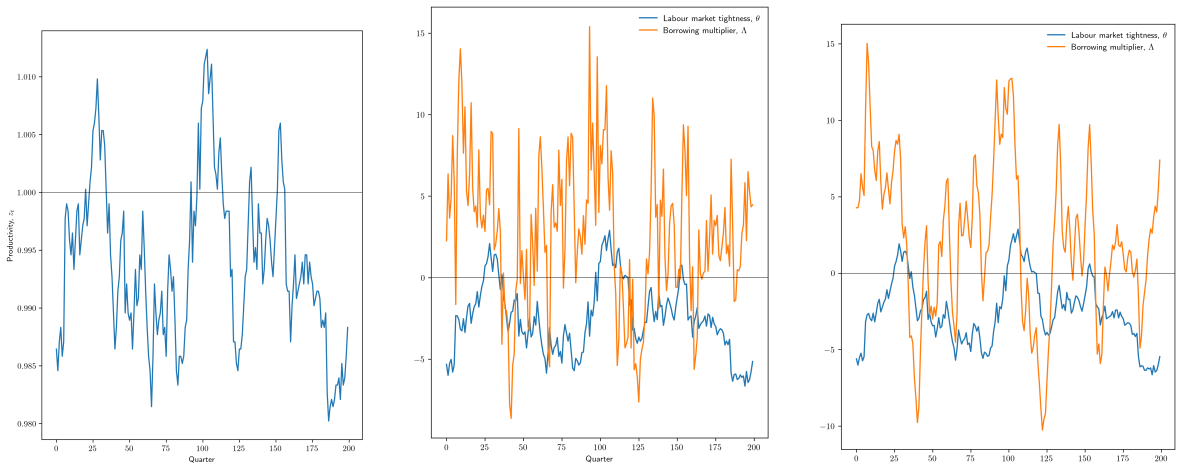
capacity to mitigate the drop in borrowing is weaker than under EBITDA constraints. Firms under EBITDA borrowing constraints can cut vacancies to maintain some cash flow during the shock, so the decline in bond sales is smaller.

5.3 Qualitative Labour–Finance Comovement in Simulated Cycles

This section compares the qualitative consistency of the labour-finance comovement in the data with the comovement generated by the EBITDA and KM economies. In particular, the following analysis tests whether the EBITDA and KM economies reproduce the sign and timing of the relation between labour-market and financial conditions documented in Figure 1 over a simulated sequence of productivity and borrowing-constraint shocks. The analysis again uses Λ as the model's counterpart to the credit tightness measure presented earlier.

As before, the EBITDA and KM economies are set with $\kappa = 0.03$, and the productivity and borrowing constraint shocks are simulated separately. To compute the simulations, the productivity and borrowing constraint shocks are discretised, with the same persistence as before, and an unconditional standard deviation of 1% in logs, into a first-order Markov chain using the Rouwenhorst (1995) method. The simulated sequences of z, ξ and ξ^{KM}, over a 10,000-period time horizon, are fed into the model's first-order impulse responses to construct the implied paths of θ and Λ.

Productivity Shock Simulation Figure 6 presents the productivity-driven fluctuations.

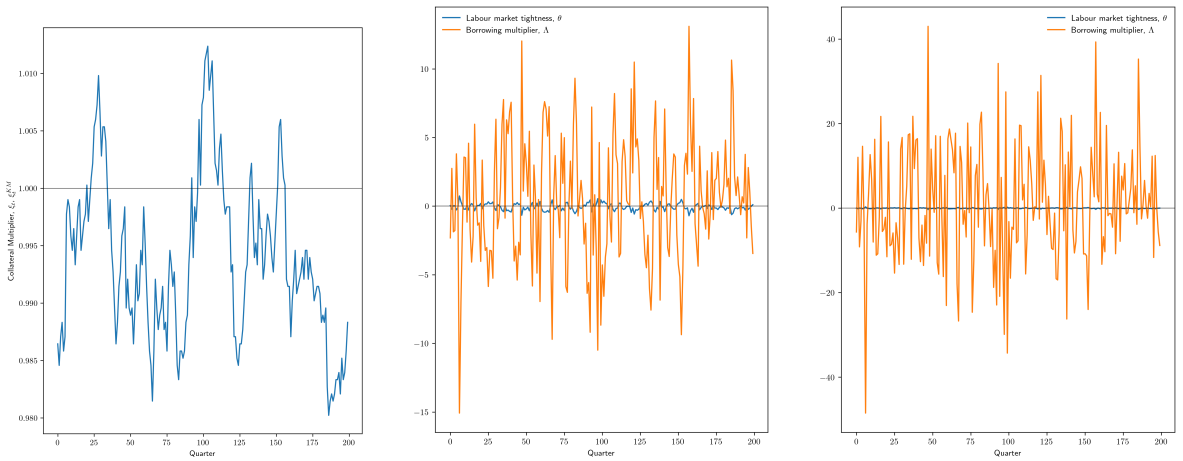


(a) z Path (b) EBITDA economy (c) KM economy

Figure 6: Percentage deviations from the steady states over z's simulated path.

Following the sequence of productivity shocks, both economies show a negative relation between θ and Λ. Considering $\Delta \log \theta$ and $\Delta \log \Lambda$, like in Figure 1, the EBITDA economy produces a contemporaneous correlation of -0.72, while the KM economy generates 0.70. The EBITDA economy, following the simulated productivity shocks, reproduces the sign can be seen from the data.

Borrowing Constraint Shock Simulation Figure 7 plots the simulated path of ξ and ξ^{KM} and the corresponding movements in θ and Λ.



(a) ξ and ξ^{KM}'s Path (b) EBITDA economy (c) KM economy

Figure 7: Percentage deviations from the steady states over ξ and ξ^{KM}'s simulated path.

As the impulse responses show, Λ is more volatile than θ in the simulations. Both economies generate a contemporaneous correlation of -1 between $\Delta \log \theta$ and $\Delta \log \Lambda$. This is expected since a change in ξ or ξ^{KM} immediately changes a firm's borrowing capacity, inducing an immediate change in how much the borrowing constraint bites and in the firm's capacity to post vacancies.

Correlations: Leads and Lags Table 4 presents lead-lag correlations between $\Delta \log \theta$ and $\Delta \log \Lambda$, computed from the simulations. Each row shows the correlations from each model for the productivity and borrowing constraint shocks.

Table 4: Correlation between $\Delta \log \theta_t$, and $\Delta \log \Lambda$, i quarters earlier.

$\text{Corr}(\Delta \log \theta_t, \Delta \log \Lambda_{t-i})$, i	-4	-3	-2	-1	0	1	2	3	4
EBITDA, productivity	-0.049	0.024	0.184	0.712	-0.720	0.075	0.017	0.019	0.008
KM, productivity	-0.200	-0.223	-0.273	0.474	0.699	0.031	0.039	0.029	0.025
EBITDA, collateral	0.013	0.052	0.107	0.335	-1.000	0.321	0.102	0.050	0.012
KM, collateral	-0.008	0.012	-0.029	0.520	-1.000	0.531	-0.031	0.011	-0.009

While the EBITDA and KM economies do not reproduce the lead-lag pattern observed in Table 1, under productivity shocks, it is only the EBITDA economy that reproduces

a contemporaneous correlation between $\Delta \log \theta$ and $\Delta \log \Lambda$ that is similar to the data. Under borrowing constraint shocks, by contrast, both economies consistently predict a negative contemporaneous correlation between the two measures. Borrowing constraints, therfore, allow the model to produce a contemporaneous correlation between changes in labour and credit tightness measures that is observed in the data following borrowing constraint shocks. Following productivity shocks, however, the EBITDA economy with dividend adjustment costs qualitatively aligns more closely with the data.

6 The Fundamental Surplus Matters

This section examines how the fundamental surplus, the quantity of resources that the invisible hand can allocate to vacancy creation (Ljungqvist and Sargent, 2017), amplifies the labour-finance transmission channel in the EBITDA borrowing-constrained economy following productivity and borrowing constraint shocks.

6.1 The Fundamental Surplus under the EBITDA Borrowing Constraint in the Steady State

Before proceeding to examine the productivity and borrowing constraint impulse responses, under different fundamental surpluses, let us consider the impact of the financial frictions on the fundamental surplus.

Taking the marginal product of capital in the steady state,

$$zF_k(k,n) = z\alpha k^{\alpha-1} n^{1-\alpha} = \frac{\beta^{-1} - (1-\delta)}{1 + \xi \frac{\Lambda}{1+\hat{\mu}}}$$

the production function's decomposition,

$$y = n \cdot zF_n(k, n) + k \cdot zF_k(k, n)$$

and the Ljungqvist and Sargent (2017) fundamental surplus definition,

$$\text{FS} = zF_n(k, n) - b$$

the fundamental surplus can be written as,

$$\text{FS} = z^{\frac{1}{1-\alpha}} \left[\frac{1 + \xi \frac{\Lambda}{1+\hat{\mu}}}{\beta^{-1} - (1-\delta)} \right]^{\frac{\alpha}{1-\alpha}} \alpha^{\frac{\alpha}{1-\alpha}} (1-\alpha) - b \tag{24}$$

A binding borrowing constraint, $\Lambda > 0$, ceteris paribus, raises the marginal product of labour through a higher capital-to-labour ratio, and thus raises the fundamental surplus. This is similar to the case where $\hat{\mu} < 0$ in the steady state: a larger capital-labour ratio raises the fundamental surplus through the same channel.

In the steady state, a low fundamental surplus reduces the firm's borrowing capacity. Using the steady state wage, the EBITDA borrowing constraint, the production function's decomposition into its marginal products, the free-entry condition and the matching function, the firm's bond sales can be written as,

$$\frac{x}{\tilde{R}} = \xi \frac{\Upsilon \theta^{1-\eta}}{s + \Upsilon \theta^{1-\eta}} \left[\frac{\alpha}{1-\alpha} (\text{FS} + b) + \frac{\gamma \rho}{\Upsilon} \theta^{\eta} \right] \tag{25}$$

(25) shows that when the fundamental surplus is low, the firm has lower EBITDA cash flow, and it can borrow less.

6.2 Quantitative Analysis

Let us now consider the effect of the fundamental surplus on the labour and financial markets following productivity and borrowing constraint shocks, in the EBITDA borrowing constraint economy. The following analysis focuses on two versions of the model: a low fundamental surplus version where b is 0.9, and a high fundamental surplus version where b is 0.4. For the rest of the parameters in both models, the same calibration strategy is used as before. The same steady state unemployment and labour market tightness targets remain the same, ϕ is also set to equal the respective steady state dividends, and the remaining parameters remain the same as before.

Regarding the effect that dividend adjustment costs have on the dynamics, the following analysis also considers the cases of $\kappa = 0$ and $\kappa = 0.1$ for both high and low fundamental surplus economies. Setting $\kappa = 0.1$ illustrates how a large adjustment cost changes the dynamics across different fundamental surpluses.

6.2.1 Productivity Shock

Beginning with a positive productivity shock, Figure 8 shows the goods market, labour and financial market responses under the two fundamental surpluses.

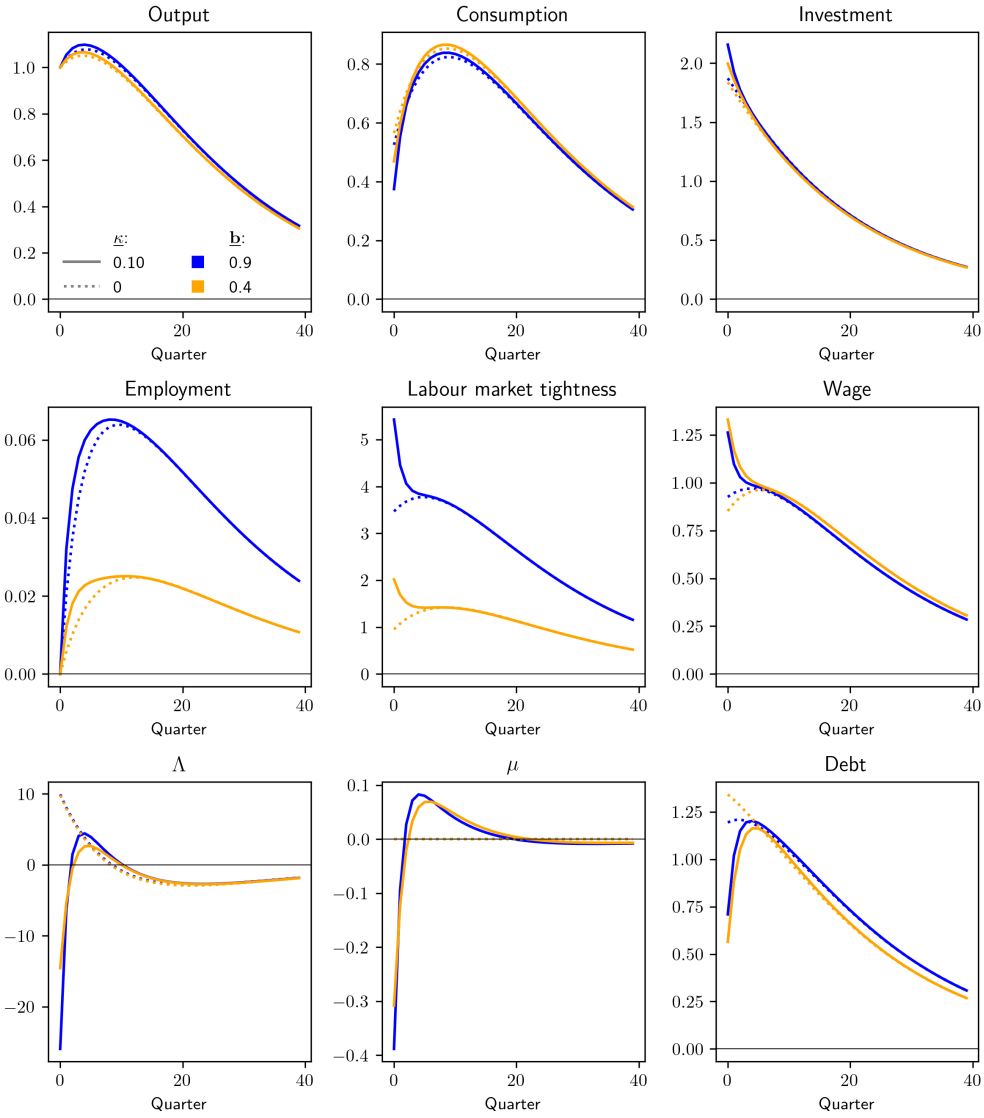


Figure 8: Low vs. High Fundamental Surplus: Positive 1% Productivity Shock. Percentage deviation from the steady state.

As expected from Ljungqvist and Sargent (2017), θ's increase following the positive productivity shock is larger when the fundamental surplus is low. In fact, between the $b = 0.9$ and $b = 0.4$ economies, the θ increase in the former is more than twice larger than in the latter. In addition, under a lower fundamental surplus there is a slight increase in output and investment responses to the shock, and greater amplification in changes in Λ and μ. The low fundamental surplus, under dividend adjustment costs, causes a larger

decline in Λ upon impact by approximately 80%, and an approximately 30% larger decline in μ upon impact. The larger workforce expansion increases borrowing capacity through higher cash flow, loosening the borrowing constraint and pushing dividends further above target.

With no dividend adjustment costs, the productivity shock does not transmit to the financial market as much. The response of Λ and debt is minimally affected by different fundamental surpluses.

6.2.2 Borrowing Constraint Shock

Figure 9 repeats the experiment for a -1% borrowing-constraint shock and reports the same variables as before.

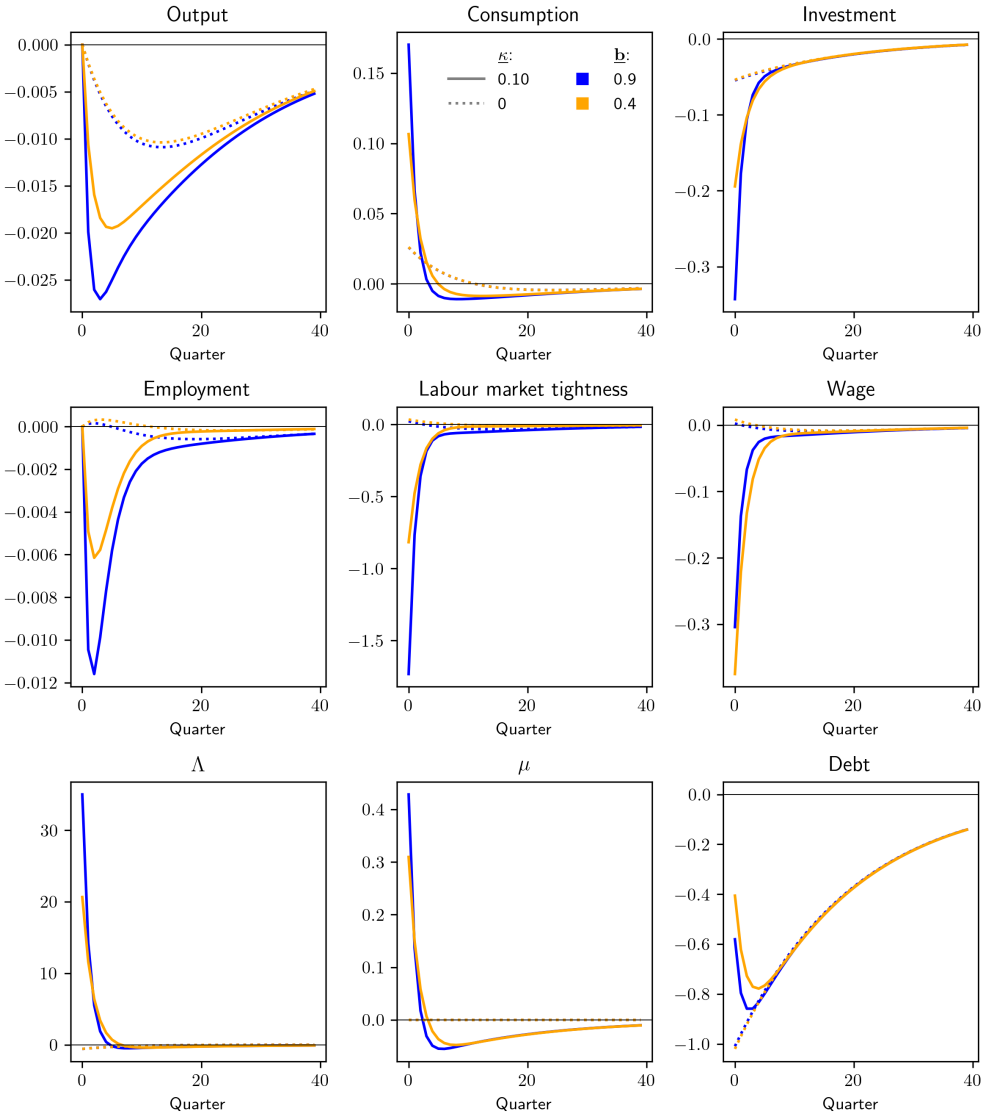


Figure 9: Low vs. High Fundamental Surplus: Negative 1% Borrowing Constraint Shock. Percentage deviation from the steady state.

The figure shows that, with dividend adjustment costs, a low fundamental surplus amplifies responses in the labour, goods, and financial markets, particularly at or near the shock's impact. With no dividend adjustment costs, however, the shock does not transmit into the labour market. With $\kappa > 0$, peak output is amplified by approximately 40% between the $b = 0.4$ and $b = 0.9$ economies, with much larger amplifications in labour-market tightness and borrowing-constraint tightness. This mechanism works

through the EBITDA borrowing constraint and the costly dividend adjustment. The shock reduces the firm's borrowing capacity, and the firm responds by cutting vacancies as it tries to recover EBITDA and dividends to mitigate the shock. When the fundamental surplus is low, the firm already has fewer resources to post vacancies; therefore, to recover some borrowing capacity, they cut more vacancies. As a consequence, the workforce reduces by more, leading to a larger decline in investment and spilling over into lower cash flow.

7 Conclusion

This paper shows that interactions between typical labour and financial market frictions explain the close links between the two markets. To do this, the paper develops a simple model with DMP labour-search frictions, costly dividend adjustment, and EBITDA borrowing constraints. The model shows that, in particular, the interaction between costly dividend adjustment and the EBITDA borrowing constraint, together with labour market frictions, brings the model closer to the labour-financial market relation observed in the data than an asset-based borrowing constraint. In addition, the model shows that, through interactions among these frictions, a low fundamental surplus amplifies transmission between labour and financial markets.

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A The Firm's Problem

Let us begin with the firm's optimisation problem,

$$J(\omega;S) = \max_{d,v,k',x'} \Bigg\{ d + \mathbb{E}\left[m' J(\omega';S')\right] + \mu\left[y - wn - \gamma v + \frac{x'}{\widetilde{R}} - x - k' + (1-\delta)k - \psi(d)\right] + \Lambda\left[\xi\left(y - wn - \gamma v\right) - \frac{x'}{\widetilde{R}}\right] \Bigg\}$$

Dividend optimality condition The first-order condition with respect to d is

$$0 = 1 - \mu\psi'(d)$$

which rearranges to (11).

Vacancy optimality condition, and value of an additional worker Turning to the first order condition with respect to vacancies,

$$0 = \mathbb{E}\left[m' \frac{\partial J(\omega';S')}{\partial v}\right] - \gamma\mu - \Lambda\xi\gamma$$

Since $\frac{\partial J(\omega';S')}{\partial v} = \frac{\partial J(\omega';S')}{\partial n'} \times \frac{\partial n'}{\partial v}$, and since $\partial n'/\partial v = q$, writing $J_n(\omega';S') = \frac{\partial J(\omega';S')}{\partial n'}$, and knowing that $\mu = 1 + \hat{\mu}$ from the main paper, the above may be written as,

$$0 = q\mathbb{E}\left[m' J_n(\omega';S')\right] - \gamma(1+\hat{\mu}) - \Lambda\xi\gamma$$

therefore,

$$q\mathbb{E}\left[m' J_n(\omega';S')\right] = \gamma(1 + \hat{\mu} + \Lambda\xi)$$

It also means that taking the envelope condition with respect to labour, and knowing from the law of motion of labour that $\partial n'/\partial n = 1 - s$,

$$J_n(\omega; S) = (1 + \hat{\mu})[zF_n(k, n) - w] + \xi\Lambda[zF_n(k, n) - w] + (1 - s)\mathbb{E}\left[m' J_n(\omega'; S')\right]$$

which can be written as,

$$J_n(\omega; S) = (1 + \hat{\mu} + \xi\Lambda)[zF_n(k, n) - w] + \xi\Lambda[zF_n(k, n) - w] + (1 - s)\mathbb{E}\left[m' J_n(\omega'; S')\right]$$

This is the value of an additional worker in the main paper.

Capital optimality condition Turning to the first-order condition with respect to capital, k', letting $J_k(\omega; S) = \partial J(\omega; S)/\partial k$,

$$0 = \mathbb{E}\left[m' J_k(\omega'; S')\right] - (1 + \hat{\mu})$$

To determine $J_k(\omega'; S')$, take the envelope condition with respect to k,

$$J_k(\omega; S) = (1 + \hat{\mu})\left[zF_k(k, n) + (1 - \delta)\right] + \xi\Lambda zF_k(k, n)$$

This means, we can write the new capital optimality condition as,

$$\mathbb{E}\left[m' \left\{(1 + \hat{\mu}')\left[z' F_k(k', n') + (1 - \delta)\right] + \xi'\Lambda' z' F_k(k', n')\right\}\right] = 1 + \hat{\mu}$$

Capital - Labour ratio in the steady state In the steady state, $m = 1/R = \beta = (1+\rho)^{-1}$, and $\hat{\mu} = \hat{\mu}'$, which turns the capital optimality condition as,

$$\frac{1}{1+\rho}\left\{(1+\hat{\mu})[zF_k(k,n)+(1-\delta)]+\xi\Lambda zF_k(k,n)\right\} = 1+\hat{\mu}$$

This can we re-written as,

$$zF_k(k,n)+(1-\delta)+\frac{\xi\Lambda}{1+\hat{\mu}}zF_k(k,n) = 1+\rho$$

therefore,

$$zF_k(k,n)\left[1+\xi\frac{\Lambda}{1+\hat{\mu}}\right] = \rho+\delta$$

Since $zF_k(k,n) = z\alpha k^{\alpha-1}n^{1-\alpha}$, then,

$$z\alpha\left(\frac{k}{n}\right)^{\alpha-1}\left[1+\xi\frac{\Lambda}{1+\hat{\mu}}\right] = \rho+\delta$$

which can be re-arranged to,

$$\frac{k}{n} = \left[\frac{\alpha z\left(1+\xi\frac{\Lambda}{1+\hat{\mu}}\right)}{\rho+\delta}\right]^{\frac{1}{1-\alpha}}$$

Debt optimality condition The first order condition with respect to x' is,

$$0 = \mathbb{E}\left[m'\frac{\partial J(\omega';S')}{\partial x'}\right]+\frac{1+\hat{\mu}}{\tilde{R}}-\frac{\Lambda}{\tilde{R}}$$

Computing the envelope condition with respect to debt,

$$\frac{\partial J(\omega;S)}{\partial x} = -(1+\hat{\mu})$$

which means the optimality can be written as,

$$\frac{\Lambda}{\tilde{R}} + \mathbb{E}\left[m'(1+\hat{\mu}')\right] = \frac{1+\hat{\mu}}{\tilde{R}}$$

B Dynamic Nash Wage Derivation

Recall that the household's Bellman equation is,

$$W(a,n;S) = \max_{c,a'} \left\{u(c) + \beta \mathbb{E}\left[W(a',n';S')\right]\right\}$$

subject to (19) and (20). We can compute the envelope condition with respect to labour as,

$$\frac{\partial W(a,n;S)}{\partial n} = W_n(a,n;S) = u'(c)(w-b) + \beta \frac{\partial n'}{\partial n} \mathbb{E}\left[W_n(a',n';S')\right]$$

Since $\partial n'/\partial n = 1-s-f$, this means,

$$W_n(a,n;S) = u'(c)(w-b) + \beta(1-s-f)\mathbb{E}\left[W_n(a',n';S')\right] \tag{26}$$

Taking the first-order condition of (22) with respect to w gives us,

$$\frac{\Psi}{1-\Psi} \frac{J_n(\omega;S)}{1+\hat{\mu}+\xi\Lambda} = \frac{W_n(a,n;S)}{u'(c)} \tag{27}$$

where we divide the firm's side by $1+\hat{\mu}+\xi\Lambda$ because $\partial J_n/\partial w = -(1+\hat{\mu}+\xi\Lambda)$. We can re-write (26) as,

$$\frac{W_n(a,n;S)}{u'(c)} = (w-b) + \beta(1-s-f)\mathbb{E}\left[\frac{u'(c')}{u'(c)} \frac{W_n(a',n';S')}{u'(c')}\right]$$

Since from the Euler equation $u'(c) = \beta R \mathbb{E}[u'(c')]$,

$$\frac{W_n(a,n;S)}{u'(c)} = (w-b) + \frac{1}{R}(1-s-f)\mathbb{E}\left[\frac{W_n(a',n';S')}{u'(c')}\right]$$

Which we can rewrite as,

$$\frac{\Psi}{1-\Psi}\frac{J_n(\omega;S)}{1+\hat{\mu}+\xi\Lambda} = (w-b) + \frac{1}{R}(1-s-f)\frac{\Psi}{1-\Psi}\mathbb{E}\left[\frac{J_n(\omega',S')}{1+\hat{\mu}'+\xi'\Lambda'}\right] \tag{28}$$

Now we can take (14), which recall is,

$$J_n(\omega;S) = (1+\hat{\mu}+\xi\Lambda)\left[zF_n(k,n) - w\right] + (1-s)\mathbb{E}\left[m' J_n(\omega';S')\right]$$

and the free-entry condition (13), which is,

$$q\mathbb{E}\left[m' J_n(\omega',S')\right] = \gamma(1+\hat{\mu}+\xi\Lambda)$$

which, since $m' = \beta u'(c')/u'(c) = 1/R$, lets us rewrite the value of an additional worker as,

$$J_n(\omega;S) = (1+\hat{\mu}+\xi\Lambda)\left[zF_n(k,n) - w\right] + (1-s)\frac{\gamma(1+\hat{\mu}+\xi\Lambda)}{q} \tag{29}$$

Substituting (29) and (13) into (28),

$$\frac{\Psi}{1-\Psi}\frac{1}{(1+\hat{\mu}+\xi\Lambda)}\left[(1+\hat{\mu}+\xi\Lambda)\left[zF_n(k,n)-w\right] + (1-s)\frac{\gamma(1+\hat{\mu}+\xi\Lambda)}{q}\right] = (w-b)+(1-s-f)\frac{\Psi}{1-\Psi}\mathbb{E}\left[\frac{m' J_n(\omega';S')}{(1+\hat{\mu}'+\xi'\Lambda')}\right]$$

Multiplying the LHS and RHS by $1-\Psi$,

$$\Psi z F_n(k,n) - \Psi w + \frac{\Psi(1-s)\gamma}{q} = w - \Psi w - (1-\Psi)b + (1-s-f)\Psi\mathbb{E}\left[\frac{m' J_n(\omega';S')}{1+\hat{\mu}'+\xi'\Lambda'}\right]$$

making the wage the subject,

$$w = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} \right] + (1-\Psi)b - (1-s-f)\Psi \mathbb{E}\left[\frac{m' J_n(\omega'; S')}{1+\hat{\mu}'+\xi'\Lambda'} \right] \tag{30}$$

Since $q\mathbb{E}\left[m' J_n(\omega'; S')\right] = \gamma(1+\hat{\mu}+\xi\Lambda)$, then, the above becomes,

$$w = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} \right] + (1-\Psi)b - (1-s-f)\Psi \frac{\gamma}{q} \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right]$$

this can be simplified to,

$$w = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} - (1-s-f)\frac{\gamma}{q} \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right] \right] + (1-\Psi)b$$

which can be written as,

$$w = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} - (1-s)\frac{\gamma}{q} \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right] + f\frac{\gamma}{q} \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right] \right] + (1-\Psi)b$$

Factorising $(1-s)\gamma/q$ and knowing that $\theta = f/q$ allows us to obtain,

$$w = \Psi \left[zF_n(k,n) + \frac{(1-s)\gamma}{q} \left[1 - \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right] \right] + \gamma\theta \mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'} \right] \right] + (1-\Psi)b$$

which is the dynamic Nash wage.

C Nash wage in the stationary equilibrium

In the steady state $\mathbb{E}\left[\frac{1+\hat{\mu}+\xi\Lambda}{1+\hat{\mu}'+\xi'\Lambda'}\right] = 1$, then,

$$w = \Psi \left[zF_n(k,n) + \gamma\theta \right] + (1-\Psi)b$$

D Deriving the Fundamental Surplus

In the steady state, (29) and the free-entry condition (13) give $q\beta J_n = \gamma(1+\hat{\mu}+\xi\Lambda)$ with $J_n = (1+\hat{\mu}+\xi\Lambda)[zF_n(k,n)-w]/(1-\beta(1-s))$. The factor $(1+\hat{\mu}+\xi\Lambda)$ appears on both sides and cancels, leaving the free-entry condition of the frictionless economy,

$$\frac{(r+s)\gamma}{q} = zF_n(k,n) - w$$

An analogue to the steady state Nash wage,

$$w = b + \frac{\Psi}{1-\Psi}(r+s+f)\left(\frac{\gamma}{q}\right)$$

Combining the two and rearranging gives us,

$$\frac{(r+s)\gamma}{q} = zF_n(k,n) - b - \frac{\Psi}{1-\Psi}(r+s+f)\frac{\gamma}{q}$$

which, after rearranging, gives

$$FS = zF_n(k,n) - b = \frac{\gamma}{q}\left[r+s+\frac{\Psi}{1-\Psi}(r+s+f)\right]$$

which is the fundamental surplus.

To compute the elasticity of labour market tightness to productivity we differentiate

$$FS = zF_n - b = \frac{\gamma}{q(\theta)}\left[r+s+\frac{\Psi}{1-\Psi}(r+s+\theta q(\theta))\right]$$

with respect to z, knowing that $q(\theta) = \Upsilon\theta^{-\eta}$, means that at the steady state,

$$\eta_{\theta,z} = \frac{r+s+\Psi\theta q}{\eta(r+s)+\Psi\theta q} \cdot \frac{zF_n(k,n)}{zF_n(k,n)-b}$$

The second factor is the Ljungqvist and Sargent (2017) inverse-fundamental-surplus term.

E A Frictionless RBC–DMP Baseline

This section sets out the frictionless baseline economy.

Households The household's problem remains the same as before, but there is no tax. This means that the household's budget constraint is,

$$c + \frac{a'}{R} = wn + (1-n)b + a + d$$

and the Euler equation remains the same, $u'(c) = R\beta\mathbb{E}[u'(c')]$.

Firms Firms do not confront a borrowing constraint. The firm solves,

$$\begin{aligned} J(\omega;S) = \max_{d,v,k',x'} \Bigg\{ & d + \mathbb{E}\left[m' J(\omega';S')\right] \\ & + \mu\left[y - wn - \gamma v + \frac{x'}{R} - x - k' + (1-\delta)k - \psi(d)\right] \Bigg\} \end{aligned}$$

The optimality conditions yield, where $\mu = 1 + \hat{\mu}$ as before. With regards to dividends,

$$1 = (1+\hat{\mu})\psi'(d)$$

with respect to vacancies,

$$q\,\mathbb{E}[m' J_n(\omega'; S')] = \gamma(1+\hat{\mu})$$

and the envelope condition with respect to labour is,

$$J_n(\omega; S) = (1+\hat{\mu})\left[zF_n(k,n) - w\right] + (1-s)\mathbb{E}\left[m' J_n(\omega'; S')\right]$$

with respect to capital,

$$\mathbb{E}\left[m'\left\{(1+\hat{\mu}')\left(z'F_k(k',n') + (1-\delta)\right)\right\}\right] = 1+\hat{\mu}$$

with respect to debt,

$$\mathbb{E}\left[m'(1+\hat{\mu}')\right] = \frac{1+\hat{\mu}}{R}$$

Nash wages With dividend adjustment costs, the Nash wage becomes,

$$w^{\text{Nash}} = \Psi\left[zF_n(k,n) + \frac{(1-s)\gamma}{q} - (1-s-f)\frac{\gamma}{q}\mathbb{E}\left[\frac{1+\hat{\mu}}{1+\hat{\mu}'}\right]\right] + (1-\Psi)b$$

Recursive Competitive Equilibrium The recursive competitive equilibrium in this model is defined similarly to the one in the main body. To close the model, $X' = K'$ is imposed to pin down the quantity of debt, thus ensuring that the aggregate resource constraint $Y(S) + (1-N)b - \gamma V(S) = C(S) + K'(S) - (1-\delta)K + \kappa(D(S)-\phi)^2$ holds.

F Kiyotaki-Moore Borrowing Constrained Economy

This section presents the Kiyotaki-Moore borrowing constrained economy.

Households. The households set up remains the same as in the main paper.

Firms. Firms confront the borrowing constraint,

$$\frac{x'}{\widetilde{R}} \leq \xi^{KM}(1-\delta)k'.$$

and solve the problem,

$$\begin{aligned} J(\omega;S) = \max_{d,v,k',x'} \Bigg\{ & d + \mathbb{E}\left[m' J(\omega';S')\right] \\ & + \mu \left[y - wn - \gamma v + \frac{x'}{\widetilde{R}} - x - k' + (1-\delta)k - \psi(d) \right] \\ & + \Lambda \left[\xi^{KM}(1-\delta)k' - \frac{x'}{\widetilde{R}} \right] \Bigg\} \end{aligned}$$

Using the same notation that $\mu \equiv 1 + \hat{\mu}$, we can write the first order condition with respect to dividends as,

$$1 = (1+\hat{\mu})\psi'(d)$$

The optimality condition with respect to vacancies is,

$$q\mathbb{E}\left[m' J_n(\omega';S')\right] = \gamma(1+\hat{\mu})$$

where the value of an additional worker is,

$$J_n(\omega;S) = (1+\hat{\mu})\left[zF_n(k,n) - w\right] + (1-s)\mathbb{E}\left[m' J_n(\omega';S')\right]$$

Capital investment's optimality condition is,

$$\mathbb{E}\left[m' J_k(\omega';S')\right] + \Lambda\xi^{KM}(1-\delta) = 1 + \hat{\mu}$$

where capital's envelope condition is,

$$J_k(\omega; S) = (1 + \hat{\mu}) \left[z F_k(k, n) + 1 - \delta \right]$$

Combining the two yields,

$$\mathbb{E}\left[m' \Big\{ (1 + \hat{\mu}') \left[z' F_k(k', n') + 1 - \delta \right] \Big\} \right] + \Lambda \xi^{KM} (1 - \delta) = 1 + \hat{\mu}$$

Debt's optimality condition can be computed by combining the first order condition,

$$\mathbb{E}\left[m' \frac{\partial J(\omega'; S')}{\partial x'} \right] + \frac{1 + \hat{\mu}}{\widetilde{R}} = \frac{\Lambda}{\widetilde{R}}$$

and the envelope condition,

$$\frac{\partial J(\omega; S)}{\partial x} = -(1 + \hat{\mu})$$

therefore,

$$\frac{\Lambda}{\widetilde{R}} + \mathbb{E}\left[m'(1 + \hat{\mu}') \right] = \frac{1 + \hat{\mu}}{\widetilde{R}}$$

Nash Wage The Nash wage is the same as in the baseline with dividend adjustment costs,

$$w^{\text{Nash}} = \Psi \left[z F_n(k, n) + \frac{(1 - s)\gamma}{q} - (1 - s - f) \frac{\gamma}{q} \mathbb{E}\left[\frac{1 + \hat{\mu}}{1 + \hat{\mu}'} \right] \right] + (1 - \Psi) b$$

Recursive Competitive Equilibrium The recursive competitive equilibrium is defined in a similar manner as in the main text.